

Dynamics: $\dot{x}(t) = v \cos \theta(t) + w(y(t))$
 $\dot{y}(t) = v \sin \theta(t)$

Constraints: $(x(t_0), y(t_0)) = (x_0, y_0)$
 $(x(t_f), y(t_f)) = (0, 0)$

Cost: $J = t_f - t_0 = \int_{t_0}^{t_f} 1 dt$

1. $w(y(t)) = \frac{V}{h} y(t)$

try to proof: $\tan \theta^*(t) = \alpha - \frac{V}{h} t$

$$H = 1 + P_x \left[v \cos \theta(t) + \frac{V}{h} y(t) \right] + P_y \cdot v \sin \theta(t)$$

$$\dot{P}_x^* = -\frac{\partial H}{\partial x} = 0$$

$$\dot{P}_y^* = -\frac{\partial H}{\partial y} = -\frac{V}{h} P_x$$

$$0 = \frac{\partial H}{\partial \theta} = -P_x V \sin \theta + P_y V \cos \theta$$

$$\therefore P_x V \sin \theta = P_y V \cos \theta$$

$$\frac{\sin \theta}{\cos \theta} = \frac{P_y V}{P_x V}$$

$$\tan \theta = \frac{P_y}{P_x}$$

$$\therefore \dot{P}_x^* = 0$$

$$\therefore P_x = C$$

$$\therefore \dot{p}_y^* = -\frac{CV}{h}$$

$$\therefore p_y = -\frac{CV}{h}t + D$$

$$\begin{aligned}\therefore \tan\theta &= \frac{p_y}{p_x} = \frac{-\frac{CV}{h}t + D}{C} = -\frac{V}{h}t + \frac{D}{C} \\ &= -\frac{V}{h}t + \alpha\end{aligned}$$

where α is a constant

$$2. \quad w(y(t)) = \beta \quad . \quad \beta > 0$$

$$\text{want: } T = t_1^* - t_0$$

$$H = 1 + p_x (v \cos\theta(t) + \beta) + p_y v \sin\theta(t)$$

$$\therefore \dot{p}_x = -\frac{\partial H}{\partial x} = 0$$

$$\dot{p}_y = -\frac{\partial H}{\partial y} = 0$$

$$\therefore p = C, \text{ constant}$$

$$0 = \frac{\partial H}{\partial \theta} = -p_x v \sin\theta + p_y v \cos\theta$$

$$\therefore p_x v \sin\theta = p_y v \cos\theta$$

$$\therefore \tan\theta = \frac{p_y}{p_x} = D, \text{ constant}$$

which means the direction is fixed

$$\begin{aligned}x(t_1) &= x(t_0) + \dot{x}(t_1) \cdot T \\ &= x_0 + (v \cos\theta + \beta) T\end{aligned}$$

$$y(t_1) = y(t_0) + \dot{y}(t_1) \cdot T$$

$$= y_0 + V \sin \theta \cdot T$$

$$\therefore \begin{cases} 0 = x_0 + (V \cos \theta + \beta) T \\ 0 = y_0 + V \sin \theta \cdot T \end{cases}$$

θ is unknown, need to be eliminated

$$\Rightarrow \begin{cases} \cos \theta = \frac{-x_0 - T\beta}{TV} \\ \sin \theta = \frac{-y_0}{TV} \end{cases}$$

$$\therefore \sin^2 \theta + \cos^2 \theta = 1$$

$$\therefore \frac{y_0^2}{T^2 V^2} + \frac{(x_0 + T\beta)^2}{T^2 V^2} = 1$$

$$y_0^2 + x_0^2 + 2x_0\beta T + T^2\beta^2 = T^2 V^2$$

$$(\beta^2 - V^2)T^2 + 2x_0\beta T + x_0^2 + y_0^2 = 0$$

$$\therefore T = \frac{-2x_0\beta \pm \sqrt{4x_0^2\beta^2 - 4(\beta^2 - V^2)(x_0^2 + y_0^2)}}{2(\beta^2 - V^2)}$$

$$= \frac{-x_0\beta \pm \sqrt{\beta^2 y_0^2 + V^2 x_0^2 + V^2 y_0^2}}{\beta^2 - V^2}$$